CSA5117

Assignment-3

Set-1

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Consider the following two hash algorithms : MD5 and SHA-1 Both are used to produce message digest

a) compare the output length of MD5 and SHA-1

MD5 and SHA-1 are cryptographic hash algorithms that produce fixed-length message digets

→ MD5 produces a 128-bit (16 byte) message digest.

→ SHA1 produces a 160-bit (20 byte) message digest.

∴ SHA-1 has a longer output than MD5.

b) Explain why MD5 is no longer consider secure.

MD5 is no longer considered secure because it has serious collision vulnerabilities . A collision occurs when two different messages produce the same hash value . Researcher have demonstrated practical methods for creating such colisions.

Because of this weakness, MD5 should not be used for security -sensitive applications such as digital signature , certifis or verifying the authenticity of important file.

c) If two different files result in the same hash in MD5. what type of attack does this represent?

If two different files produce the same MD5 hash, it represents a collision attack.

For example

File A ≠ File B

but

MD5 (File A) = MD5 (File B)

This shows the collision resistance of MD5 has been broken.

d) why is SHA-256 considered more secure than SHA-1? illustrate with reference to collision resistance.

SHA-256 is considered more secure than SHA-1 because it produces a 256 bit hash, while SHA-1 produces a 160 bit hash. A longer hash size provides stronger resistance against collision attacks.

For an ideal hash function, the approximate work required to find a collision is $2^{n/2}$ where n is the hash length.

$\Rightarrow$ SHA-1: $2^{160/2} = 2^{80}$ operation.

$\Rightarrow$ SHA-256: $2^{256/2} = 2^{128}$ operation.

thus, finding a collision in SHA-256 requires theoretically about 2^{128} operation which is far more difficult than the 2^{80} operation associated with SHA-1